

April Zhao

Java Full Stack Developer

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Summary

- 10 years of hands-on experience designing and developing enterprise-grade **Java/J2EE** applications, applying advanced object-oriented design principles and patterns.
- Proficient in building scalable microservices and **RESTful APIs** using **Spring Boot**, with expertise in containerization (**Docker**) and orchestration (**Kubernetes**).
- Strong background in **Oracle, MySQL, PostgreSQL, MSSQL, MongoDB, and Cassandra**, including SQL optimization, **PL/SQL**, and **NoSQL** modeling.
- Strong background in implementing robust data persistence layers using **Hibernate/JPA** and relational databases (**Oracle, MySQL**), including SQL tuning and stored procedures.
- Skilled in modern front-end development (**Angular, React, JavaScript, TypeScript and Node.js.**), ensuring seamless integration with back-end services.
- Experienced with application servers (**Tomcat, WebSphere, JBoss**), messaging solutions (**JMS**), and Web Services (**SOAP, REST**).
- Implemented **OAuth2, JWT, SSO (Okta)**, **HTTPS** encryption, and **CORS** for secure web applications.
- Adept in **Agile/Scrum** methodologies, **CI/CD pipelines (Jenkins)**, and **TDD/BDD** practices (**JUnit, Cucumber**) for streamlined, high-quality releases.
- Extensive experience in version control (**Git, SVN**) and **DevOps** practices, ensuring efficient build, deployment, and monitoring processes.
- Hands-on experience with **Apache Kafka, RabbitMQ, Spring JMS**, and **ActiveMQ** for event-driven architectures.
- Expertise in **AWS (EC2, S3, Lambda, RDS, ECS, EKS, IAM, ElasticSearch)** and **DevOps** practices with **Jenkins, Docker, Kubernetes, GIT**, and **Chef**.
- Experience with **Log4j, SLF4J, ELK** stack, **AWS CloudWatch**, and log aggregation for production monitoring.
- Skilled in **JIRA, Git Flow, Confluence**, and Agile sprint planning for effective team collaboration.
- A collaborative and adaptable professional with strong communication skills and a proactive mindset, eager to learn and take initiative, ensuring the successful execution of projects.

Technical Skills

Languages	Java, Python, C, C++, PL/SQL, CSS, TypeScript,
Build & Deployment	Maven, Gradle, Docker, Kubernetes
CI/CD & Code Scan	Jenkins, SonarQube
Cloud	AWS EC2, S3, Lambda, RDS, SQS, SNS, ECS, IAM
Database	Oracle DB, MySQL, SQL Server, MongoDB, Cassandra
Front End	Angular, React
Log & Monitor & Alarm	Slf4j, Log Files, Log Aggregation, ELK, Spring Actuator, AWS CloudWatch
Messaging System	JMS, Apache Kafka, RabbitMQ, AWS SQS, AWS SNS

Networking	HTTP Protocol, DNS, HTTPS, Load Balancer, Reverse Proxy, Cloud
Security	JWT, HTTPS, Encryption, CORS, OAuth2
Team Work	SDLC, BDD/TDD, Agile, Scrum, JIRA, Git Flow, Production Support
Testing	Unit Test, JUnit, Mockito, JMeter, Cucumber, Karate Lab, UAT

Experience

June 2023 – Present | New York, NY

Client Name: Bloomingdale's

Role: Senior software developer

Project: In-store & Online Inventory Management System Optimization (BOPIS)

Project Description:

Optimizing inventory tracking and stock availability across multiple store locations and online platforms. The solution improved operational efficiency, reduced stock discrepancies, and enhanced real-time inventory visibility, resulting in increased sales and revenue.

Responsibilities:

- **Designed and developed** the backend system using **Java 17** and **Spring Boot** within a **microservices architecture**, ensuring high scalability and maintainability.
- **Built and maintained** the front-end application using **ReactJS, HTML, CSS, and TypeScript**, providing an intuitive and responsive interface for store associates.
- **Developed RESTful APIs** to facilitate seamless interaction between inventory databases and front-end applications.
- **Optimized inventory processing workflows**, enhancing **real-time stock updates** and minimizing discrepancies.
- **Collaborated closely** with the **Product Manager and Engineering Manager** to define the product roadmap, set strategic goals, and ensure successful feature delivery.
- **Managed and refactored legacy codebases**, integrating **modern ReactJS infrastructure** to improve maintainability and performance.
- **Implemented robust unit testing** using **JUnit and Mockito** to ensure code quality and maintainability.
- **Utilized Git** for **version control** and enforced best practices in code reviews.
- **Automated CI/CD pipelines** using **Jenkins**, streamlining deployment processes and improving development efficiency.
- **Integrated Single Sign-On (SSO) authentication** using **Okta API** to enhance security and user management.
- **Deployed and scaled the application** on **AWS cloud infrastructure**, leveraging **EC2, ECS, ECR, RDS, and S3** for high availability and reliability.
- **Containerized applications** using **Docker** and orchestrated services using **Kubernetes** to optimize deployment efficiency.
- **Monitored and managed deployments** to ensure seamless production releases, addressing potential issues proactively.
- **Utilized JIRA** for efficient **ticket management, sprint planning, and Agile collaboration** within cross-functional teams.

Environment: Java 17, Spring Boot, ReactJS, HTML, CSS, TypeScript, AWS S3, EC2, ECS, ECR, RDS, RESTful APIs, Docker, Kubernetes, Git, Jenkins, Okta API (SSO), JIRA, IntelliJ IDEA, Visual Studio Code.

2021 November - 2023 May | New York, NY

Client Name: Verizon

Project Name: User Information Dashboard

Role: Senior Full Stack Developer

Description:

The system integrates **real-time data processing, advanced visualization, and role-based access control**, enabling **customer support teams, analysts, and executives** to make **data-driven decisions**.

Responsibilities:

- Developed and maintained the backend **microservices** using **Java 17, Spring Boot**, ensuring **efficient data retrieval and processing**.
- Built **RESTful APIs** to facilitate secure and scalable communication between the **frontend UI and backend services**.
- **Designed and optimized database schemas** in **PostgreSQL**, ensuring **efficient query execution and fast data retrieval**.
- **Developed the frontend UI** using **React, Material-UI, JavaScript, and TypeScript**, providing a **user-friendly and responsive interface**.
- **Implemented real-time data updates** using **WebSockets and Server-Sent Events (SSE)**, enhancing **live user tracking and dashboard interactivity**.
- **Integrated authentication and authorization mechanisms** using **JWT and OAuth2**, ensuring **secure access to user information**.
- Optimized API performance through **multithreading and SQL query tuning**, improving response times for large datasets.
- **Managed CI/CD pipelines** using **Jenkins**, automating build, testing, and deployment processes.
- **Containerized applications** using **Docker** and deployed services on **AWS (EC2, RDS, S3)** for scalability and high availability.
- **Utilized Git for version control and collaborated using JIRA for agile project management and sprint tracking**.

Environment: Java 17, Spring Boot, React, TypeScript, Material-UI, PostgreSQL, RESTful APIs, WebSockets, Server-Sent Events (SSE), JWT, OAuth2, Docker, AWS (EC2, RDS, S3), Jenkins, Multithreading, Git, JIRA.

2019 September - 2021 October | New York, NY

Client Name: Deutsche Bank

Project Name: Data Migration Platform for Deutsche Bank

Role: Java Backend Developer

Description:

The platform ensures transfer and integration of **large-scale financial datasets** across legacy and modern systems through **optimized ETL pipelines and microservices architecture**.

Responsibilities:

- **Developed and maintained** scalable **Java-based microservices** using **Spring Boot**, handling **data extraction, transformation, and loading (ETL) processes**.
- **Designed and implemented RESTful APIs** to facilitate secure and efficient data exchange between various financial systems and third-party services.
- **Optimized database operations**, ensuring **high-performance queries** and transaction integrity in **PostgreSQL and Oracle databases**.
- **Developed custom batch processing solutions** using **Spring Batch**, enabling efficient migration of large-scale financial data.
- **Integrated Kafka-based messaging** for real-time **event-driven data streaming** and synchronization across distributed systems.
- **Ensured data security and compliance** by implementing **JWT-based authentication, OAuth2, and role-based access control (RBAC)** for secure API access.

- **Containerized applications** using **Docker** and managed container orchestration with **Kubernetes** for scalable deployments.
- **Developed unit and integration tests** using **JUnit** and **Mockito**, ensuring **high code quality and system stability**.
- **Managed source control** and collaborated with cross-functional teams using **Git** and **JIRA** for issue tracking and agile sprint planning.
- **Utilized Hibernate ORM** to optimize database interactions and manage complex entity relationships.
- **Implemented CI/CD pipelines** with **Jenkins**, automating the build, test, and deployment process for smooth and reliable releases.

Environment: Java 17, Spring Boot, React, TypeScript, Material-UI, PostgreSQL, RESTful APIs, WebSockets, Server-Sent Events (SSE), JWT, OAuth2, Docker, AWS (EC2, RDS, S3), Jenkins, Multithreading, Git, JIRA.

2017 June - 2019 August | New York, NY

Client Name: Hungry Panda

Project: Driver Dispatch System

Role: Full Stack Developer

Description:

Designed and developed Hungry Panda's Driver Dispatch System, a platform for dynamically assigning orders to drivers based on real-time location, availability, and traffic conditions. The system automates dispatching, optimizes delivery routes, and improves order fulfillment efficiency, reducing delivery times and increasing driver productivity.

Responsibilities:

- **Contributed to all phases** of the software development life cycle, including requirement gathering, system design, development, and testing, following **Agile/SCRUM** methodologies.
- **Developed RESTful APIs** using **Spring Boot** to handle order assignments, driver status updates, route optimization, and real-time tracking within a **microservices architecture**.
- **Implemented real-time messaging** using **Apache Kafka**, improving order assignment speed and ensuring efficient communication between dispatchers, drivers, and customers.
- **Optimized database operations** by leveraging **PostgreSQL** for transactional order data, **MongoDB** for flexible storage of driver tracking data, and **Redis** for caching real-time dispatch information, reducing latency and improving response times.
- **Utilized Hibernate ORM** to streamline database interactions and manage complex queries efficiently.
- **Deployed microservices on AWS**, utilizing **EKS, EC2, RDS, and S3** for **scalable storage and computing resources**.
- **Containerized applications** using **Docker** and orchestrated deployments with **Kubernetes**, ensuring high availability and fault tolerance.
- **Implemented unit testing** with **JUnit** and **Mockito**, ensuring system stability and code quality.
- **Enhanced security** by implementing **JWT** and **OAuth2** for authentication and authorization, ensuring secure driver logins and API access.
- **Configured centralized logging** with **Log4j** for monitoring and error tracking.

Environment: Java 17, Spring Boot, ReactJS, TypeScript, HTML, CSS, PostgreSQL, MongoDB, Redis, RESTful APIs, Apache Kafka, Hibernate, AWS (EKS, EC2, RDS, S3), Docker, Kubernetes, Jenkins, JWT, OAuth2, Log4j, Git, JIRA.

2015 August – 2017 May | New York, NY

Client Name: Etsy

Project Name: Logistics System

Role: Full Stack Developer

Description:

The system streamlined inventory distribution, automated shipping label generation, and improved real-time package tracking, ensuring efficient delivery operations for both domestic and international shipments.

Responsibilities:

- Participated in requirement gathering, analysis, design, and development of the logistics management platform, ensuring optimal system performance and scalability.
- **Developed RESTful APIs** using **Spring Boot**, enabling seamless communication between **warehouse management, order tracking, and transportation modules**.
- **Designed and implemented frontend interfaces** using **React, HTML, CSS, and JavaScript**, ensuring a smooth user experience for warehouse staff and delivery partners.
- **Integrated third-party shipping APIs** for **real-time carrier tracking, shipping label generation, and cost optimization**.
- **Managed and optimized database operations** using **PostgreSQL**, writing **complex SQL queries, stored procedures, and triggers** to support high-volume transaction processing.
- **Utilized Java concurrency and collections** to improve **data processing efficiency** for high-demand logistics workflows.
- **Implemented role-based authentication and authorization** using **JWT and OAuth2**, ensuring secure access control for different user groups.
- **Developed automated batch processing systems** for scheduling inventory updates and shipment tracking.
- **Utilized Docker for containerization** and **deployed microservices on AWS (EC2, S3, RDS, and Lambda)** to achieve high availability and scalability.
- **Wrote JUnit test cases** for unit testing, ensuring **reliable and maintainable code**.
- **Configured CI/CD pipelines using Jenkins**, automating the build, test, and deployment process.
- **Monitored and debugged application performance issues** using **Log4j** and centralized logging solutions.
- **Used Git for version control and JIRA for agile sprint planning and issue tracking**.

Environment: Java 8, Spring Boot, React, JavaScript, HTML, CSS, PostgreSQL, RESTful APIs, JWT, OAuth2, AWS (EC2, S3, RDS, Lambda), Docker, Jenkins, JUnit, Log4j, Git, JIRA.

Education

M.S. in Computer Science

Monroe University, NY

M.S. in Integrated Marketing

New York University, NY

B.A. in Communication

Communication University of China, Beijing